

Collaborative Model Optimization in Federated Stochastic Process Discovery: A Tool Report

Compiled at 8:29pm on Sunday 31st May, 2026

1 Introduction

This report summarizes the setup and results of a process discovery experiment conducted using the tool prepared for the paper titled “Collaborative Model Optimization in Federated Stochastic Process Discovery.” The following section outlines the discovery techniques used and their configurations in the experiment. Section 3 specifies the evaluation criteria used to assess the quality of the discovered models and to compare the performance of the discovery techniques. Finally, Section 4 presents the experimental results.

2 Discovery

The command line used to execute the experiment is listed below:

```
-mspd -el "Hospital Billing - Event Log.xes" m1 m2 m3 -p 50 -mxItr 20 -t 3600  
-mms 1000 -pfs 100 -ebrm U - optm DFFA
```

Table 1 summarizes the key characteristics of the experimental setup used for performance comparison.

These discovery techniques are used in the experiment:

- GASPD [3];
- FedGASPD [4];
- OptFedGASPD;

3 Quality

The quality of the discovered models is assessed using two criteria: *entropic relevance*, which captures stochastic precision and recall [2], and model *size*, defined as the number of nodes and edges, which reflects model simplicity.

Table 1: Characteristics of the experiment

Parameter	Value
Population size	50
Iterations	20
Time limit (s)	3600
Pareto front size	100
Maximum model size	1000
Optimized models	DFFA
Entropic relevance background model	U
Event log	Hospital Billing - Event Log.xes
Fitness coefficient α	0.0001 0.002
Fitness Fuction	$\frac{\alpha}{size} + \frac{1-\alpha}{Er}$
Entropic relevance degradation threshold (tau)	0.09

4 Results

Figure 1 presents the discovered deterministic frequency finite automata (DFFA) models with respect to entropic relevance and model size, where lower values are preferred for both criteria.

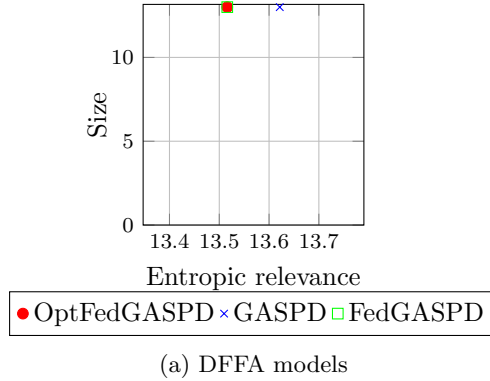


Figure 1: Discovered models plotted by size and entropic relevance for each representation type.

Table 2 reports the execution time (in iterations per second) for each input event log and method.

Table 2: Execution time comparison (iterations per second)

Event log	OptFedGASPD	GASPD	FedGASPD
Hospital Billing - Event Log.xes	362.35	539.46	362.34

References

- [1] Hootan Zhian, Rajkumar Buyya, and Artem Polyvyanyy: *Multi-Objective Metaheuristics for Effective and Efficient Stochastic Process Discovery*. Proceedings of the 23rd International Conference on Business Process Management (BPM) Seville, Spain, August 31–September 5, 2025.
- [2] Hanan Alkhamash, Artem Polyvyanyy, Alistair Moffat, Luciano García-Bañuelos: *Stochastic Directly-Follows Process Discovery Using Grammatical Inference*. Inf. Syst. 107: 101922 (2022)
- [3] Hanan Alkhamash, Artem Polyvyanyy, Alistair Moffat: *Stochastic Directly-Follows Process Discovery Using Grammatical Inference*. In: CAiSE, LNCS, vol. 14663, pp. 87–103, Springer (2024).
- [4] Hootan Zhian, Rajkumar Buyya, and Artem Polyvyanyy: *Federated Stochastic Process Discovery Using Gramatical Inference*. In: CAiSE, LNCS, vol. 15702, pp. 76–93, Springer (2025).